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Welcome to RIFRMap

An interactive platform for integrating, harmonizing, and visualizing rifampicin resistance-associated mutations across bacteria

About Rifampicin Resistance

Rifampicin (RIF) is a frontline antibiotic widely used in the treatment of tuberculosis and other bacterial infections. Resistance most commonly arises through mutations in the *rpoB* gene, which encodes the beta subunit of RNA polymerase.

These mutations alter the drug-binding site, reducing antibiotic effectiveness and contributing to the global burden of antimicrobial resistance.

Almost all resistance-associated mutations are concentrated within four regions of *rpoB*, known as clusters I, II, III, and N. These are mutational hotspots where resistance repeatedly evolves across diverse bacterial taxa.

RIFRMap enables systematic exploration of these mutations across multiple curated datasets (CARD, CRyPTIC, DRAGdb, PointFinder, Reference Gene Catalog, RifReport, and Yang data), allowing comparative analysis and deeper insights into resistance patterns.

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Data

What's new — release 2026-09-10

- **CARD** : Reprocessed with no net record-level changes detected.
- **CRyPTIC** : 139 mutation record(s) removed.
- **DRAGdb** : Reprocessed with no net record-level changes detected.
- **Yang-RIF** : Reprocessed with no net record-level changes detected.
- **RifReport** : 30 record(s) updated (AA_pos, Ref_codes).
- **PointFinder** : Reprocessed with no net record-level changes detected.
- **Reference Gene Catalog** : Reprocessed with no net record-level changes detected.

Mutation Data

Version: v1.0.0 | Last updated: 10/09/2026

Download mutations (CSV)

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Search:

Species	Gene	AA_Pos_Ecoli	AA_Pos	Original	Mutation	AA_mut_name_Ecoli	AA_mut_name	Source	Ref_codes
Bacillus subtilis	rpoB	513	469	Q	K	Q513K	Q469K	CARD	Ingham2000
Bacillus subtilis	rpoB	513	469	Q	R	Q513R	Q469R	CARD	Ingham2000
Bacillus subtilis	rpoB	522	478	A	N	A522N	A478N	CARD	Nicholson2002
Bacillus subtilis	rpoB	526	482	H	Y	H526Y	H482Y	CARD	Ingham2000
Bacillus subtilis	rpoB	531	487	S	L	S531L	S487L	CARD	Nicholson2002
Clostridioides difficile	rpoB	512	488	S	P	S512P	S488P	CARD	Miller2011
Clostridioides difficile	rpoB	512	488	S	T	S512T	S488T	CARD	OConnor2008
Clostridioides difficile	rpoB	516	492	D	N	D516N	D492N	CARD	Pecavar2012
Clostridioides difficile	rpoB	516	492	D	V	D516V	D492V	CARD	Pecavar2012
Clostridioides difficile	rpoB	520	496	P	S	P520S	P496S	CARD	Carman2012

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Reference Data

Version: v1.0.0 | Last updated: 10/09/2026

Download references (CSV)

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Search:

Ref_code	Year	Full_ref	DOI	PMID
Aiba2013	2013	Aiba Y, Katayama Y, Hishinuma T, Murakami-Kuroda H, Cui L, Hiramatsu K. Mutation of RNA polymerase beta-subunit gene promotes heterogeneous-to-homogeneous conversion of beta-lactam resistance in methicillin-resistant Staphylococcus aureus. Antimicrob Agents Chemother. 2013 Oct;57(10):4861-71. doi: 10.1128/AAC.00720-13. Epub 2013 Jul 22. PMID: 23877693; PMCID: PMC3811421.	https://doi.org/10.1128/AAC.00720-13	23877693
Alam2022	2022	Alam KM, Yan YL, Lin M et al. Rifampicin-resistant rpoB mutation in P. stutzeri enhances secondary-metabolite biosynthesis. Arch Microbiol. 2022;204:3064. doi:10.1007/s00203-022-03064-9.	https://doi.org/10.1007/s00203-022-03064-9	35768665
Albano2019	2019	Albano M, Karau MJ, Greenwood-Quaintance KE, Osmon DR, Oravec CP, Berry DJ, et al. In Vitro Activity of Rifampin, Rifabutin, Rifapentine, and Rifaximin against Planktonic and Biofilm States of Staphylococci Isolated from Periprosthetic Joint Infection. Antimicrob Agents Chemother. 2019 Oct 22;63(11):e00959-19. doi: 10.1128/AAC.00959-19. Print 2019 Nov. PMID: 31451499; PMCID: PMC6811393.	https://doi.org/10.1128/AAC.00959-19	31451499
Alekshun1997	2000	Alekshun M, Kashlev M, Schwartz I. Molecular cloning and characterization of Borrelia burgdorferi rpoB. Gene. 1997 Feb 28;186(2):227-35. doi: 10.1016/s0378-1119(96)00714-7. PMID: 9074501.	https://doi.org/10.1016/s0378-1119(96)00714-7	9074501
Alvarez2019	2019	Alvarez-Narváez S, Giguère S, Anastasi E et al. Clonal confinement of a highly mobile resistance element driven by combination therapy in Rhodococcus equi. mBio. 2019;10:e02260-19. doi:10.1128/mBio.02260-19.	https://doi.org/10.1128/mBio.02260-19	31615959
Andre2017	2017	Andre E, Goeminne L, Colmant A, Beckert P, Niemann S, Delmee M. Novel rapid PCR for the detection of Ile491Phe rpoB mutation of Mycobacterium tuberculosis, a rifampicin-resistance-conferring mutation undetected by commercial assays. Clin Microbiol Infect. 2017 Apr;23(4):267.e5-267.e7. doi: 10.1016/j.cmi.2016.12.009. Epub 2016 Dec 18. PMID: 27998822.	https://doi.org/10.1016/j.cmi.2016.12.009	27998822
Aosh2003	2003	Asoh N, Watanabe H, Fines-Guyon M, Watanabe K, Oishi K, Kositsakulchai W, Sanchai T, Kunsuikmengrai K, Kahintapong S, Khantawa B, Tharavichitkul P, Sirisanthana T, Nagatake T. Emergence of rifampin-resistant Rhodococcus equi with several types of mutations in the rpoB gene among AIDS patients in northern Thailand. J Clin Microbiol. 2003 Jun;41(6):2337-40. doi: 10.1128/JCM.41.6.2337-2340.2003. PMID: 12791846; PMCID: PMC156560.	https://doi.org/10.1128/JCM.41.6.2337-2340.2003	12791846
Aslan2008	2008	Aslan G, Tezcan S, Serin MS, Emekdas G. Genotypic analysis of isoniazid and rifampin resistance in drug-resistant clinical Mycobacterium tuberculosis complex isolates in southern Turkey. Jpn J Infect Dis. 2008 Jul;61(4):255-60. PMID: 18653964.	https://pubmed.ncbi.nlm.nih.gov/18653964/	18653964
Aubry1998	1998	Aubry-Damon H, Soussy CJ, Courvalin P. Characterization of mutations in the rpoB gene that confer rifampin resistance in Staphylococcus aureus. Antimicrob Agents Chemother. 1998 Oct;42(10):2590-4. doi: 10.1128/AAC.42.10.2590. PMID: 9756760; PMCID: PMC105902.	https://doi.org/10.1128/AAC.42.10.2590	9756760
Baek2015	2015	Baek KT, Thogersen L, Mogenssen RG, Mellergaard M, Thomsen LE, Petersen A, et al. Stepwise decrease in daptomycin susceptibility in clinical Staphylococcus aureus isolates associated with an initial mutation in rpoB and a compensatory inactivation of the clpX gene. Antimicrob Agents Chemother. 2015 Nov;59(11):6983-91. doi: 10.1128/AAC.01303-15. Epub 2015 Aug 31. PMID: 26324273; PMCID: PMC4604412.	https://doi.org/10.1128/AAC.01303-15	26324273

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